

ALLEGHENY COUNTY AP, PA, USA

WMO: 725205

Lat: 40.355N

Lon: 79.922W

Elev: 1248

StdP: 14.05

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14762

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.2	(c) 9.8	(d) -4.5	(e) 4.5	(f) 8.8	(g) -0.2	(h) 5.7	(i) 12.9	(j) 24.9	(k) 35.7	(l) 22.1	(m) 36.0	(n) 9.3	(o) 260	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.3	(c) 88.5	(d) 72.2	(e) 86.2	(f) 71.0	(g) 83.9	(h) 69.8	(i) 75.1	(j) 84.1	(k) 73.5	(l) 82.0	(m) 72.3	(n) 80.4	(o) 8.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	125.5	79.6	71.0	119.6	78.0	69.9	115.1	76.9	39.4	84.3	37.9	82.1	36.7	80.1	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.2	16.4	0.7	91.7	7.6	3.2	-4.7	94.0	-9.2	95.9	-13.4	97.7	-19.0	100.0		
			-0.3	77.8	7.2	2.6	-5.5	79.7	-9.7	81.2	-13.8	82.7	-19.1	84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	29.3	32.1	40.4	52.5	61.9	69.8	73.4	72.1	65.7	54.5	43.4	34.5
	DBStd	17.95	12.38	11.45	11.30	9.74	8.11	6.03	4.78	4.78	6.78	8.72	9.52	10.27
	HDD50	2387	653	512	344	85	7	0	0	0	0	53	241	492
	HDD65	5385	1108	921	766	388	159	26	3	5	72	343	648	946
	CDD50	3325	10	10	45	160	376	593	726	687	472	191	44	11
	CDD65	847	0	0	2	12	63	169	263	227	94	17	0	0
	CDH74	6255	0	1	13	123	520	1251	1998	1588	660	98	3	0
	CDH80	1681	0	0	1	17	110	343	596	438	163	13	0	0
Wind	WSAvg	7.8	9.5	9.1	9.0	8.9	7.5	7.0	6.4	5.9	6.1	7.3	8.3	8.9
Precipitation	PrecAvg	40.0	3.3	2.5	3.3	3.4	3.8	4.1	4.0	3.6	3.4	2.7	2.8	3.2
	PrecMax	51.3	7.1	5.0	6.1	5.2	6.1	5.8	8.0	7.8	8.8	5.5	5.9	7.1
	PrecMin	33.2	1.7	1.0	1.5	1.5	1.5	1.4	2.2	1.5	1.3	1.1	0.7	1.9
	PrecStd	5.1	1.3	0.9	1.3	0.9	1.1	1.3	1.3	1.5	1.6	1.1	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	67.2	75.0	82.4	86.5	90.4	91.4	91.1	88.4	81.6	71.7	64.2
		MCWB	56.0	53.7	58.8	62.4	69.3	72.4	74.8	73.5	70.0	67.2	59.5	56.2
	2%	DB	58.7	60.6	69.9	77.8	83.0	86.7	88.5	87.8	84.4	76.6	66.3	59.4
		MCWB	53.8	51.7	55.9	59.5	66.8	71.4	72.9	72.3	69.0	63.7	55.6	53.9
	5%	DB	53.6	54.8	63.9	73.2	80.2	84.0	85.8	84.5	81.2	72.7	62.8	54.8
		MCWB	49.7	47.4	52.3	57.9	65.7	70.3	71.3	70.7	68.2	61.6	53.9	49.6
	10%	DB	46.8	48.9	58.2	68.9	76.4	81.3	83.3	81.9	77.7	68.7	58.6	50.1
		MCWB	42.6	42.7	49.6	55.5	63.6	68.8	70.3	69.6	66.0	60.2	51.7	45.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	57.4	60.6	65.4	72.3	75.4	77.8	77.0	74.1	70.5	62.0	58.7
		MCDB	62.5	64.1	72.0	77.2	83.0	85.5	87.8	86.1	83.2	77.8	68.6	62.6
	2%	WB	54.3	53.5	57.4	62.5	69.5	73.5	75.5	74.6	72.0	67.1	58.5	54.7
		MCDB	57.8	58.8	66.0	73.0	79.4	83.0	84.6	82.9	79.7	73.6	64.1	58.4
	5%	WB	49.5	48.0	54.6	60.2	67.6	72.0	73.8	73.1	70.1	64.0	55.5	50.6
		MCDB	53.4	54.4	62.3	69.3	76.4	81.2	82.1	80.9	76.7	69.9	60.8	54.2
	10%	WB	43.0	43.2	50.3	57.9	65.5	70.5	72.3	71.6	68.2	60.8	52.4	45.6
		MCDB	46.6	48.6	57.7	66.3	73.7	78.7	80.1	79.0	74.9	67.5	57.8	49.0
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.8	17.1	19.7	18.7	17.5	17.3	17.2	17.7	16.8	14.8	12.6
		MCDBR	18.5	22.8	23.8	24.5	22.7	20.6	19.8	20.2	21.0	21.7	20.5	18.9
		MCWBR	15.0	17.2	14.3	13.0	10.4	8.9	7.7	7.9	8.7	11.1	13.7	15.3
	5% WB	MCDBR	17.8	21.2	21.3	21.1	19.4	18.0	17.0	16.7	17.2	17.8	17.9	18.0
		MCWBR	15.5	17.6	14.6	13.2	10.0	9.0	8.2	8.0	8.3	10.8	14.2	16.4
Clear-Sky Solar Irradiance	taub		0.303	0.325	0.350	0.380	0.425	0.457	0.467	0.466	0.389	0.363	0.319	0.299
	taud		2.458	2.376	2.385	2.330	2.215	2.160	2.149	2.135	2.332	2.406	2.510	2.512
	Ebn at Noon		273	282	286	283	272	262	258	254	269	264	266	266
	Edh at Noon		25	31	35	39	45	47	47	47	36	30	23	22
All-Sky Solar Radiation	RadAvg		482	747	1057	1436	1664	1811	1817	1645	1346	904	615	409
	RadStd		47	107	100	138	142	147	119	97	140	92	80	44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	N/A
Regional (37 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+104

Nomenclature: See separate page